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JT8D-219 LPT Blade 3rd & 4th, Engine Operation and Performance Approval Test Plan

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0 General

0.1 Amendment Record

#	Issue	Revision	Amendment Description	Date
1	01	00	Initial	**2025-05-22**

0.2 Affected Engines

Type/Model	SN	TC No./TCDS No.	Manufacturer
JT8D-217C, JT8D-219	-	E9NE	Pratt & Whitney (P&W)

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1 Project

1.1 Engine Type/Model

- JT8D-200 series engine

1.2 Project No.

- GT-25

2 Certification Specification

The type certification basis consists of the latest applicable airworthiness code effective on the date of the application of CP. Since this project is a major design project based on the classification Doc. No. AEC_201337_0A, the latest version of the certification specification (14 CFR Part 33-2022) should be used.

3 General Part Information

This project has been raised since the FAA issued an Airworthiness Directive (AD) 2021-22-19 for all Pratt & Whitney (P&W) JT8D-217C, and JT8D-219 turbofan engines. This AD requires operating companies to replace the rotary blades of the third and fourth stages of low pressure turbine after 5,000 hours of operation. Therefore, this project has been defined by the order of customer (one of the airlines) to implement the mentioned AD issued by FAA on the LPT blades of JT8D engines.

Therefore, the scope of this project is to design and manufacture the 3rd and 4th stages rotary blades of the low pressure turbine of the mentioned JT8D engines and testing according to the related standards.

The test equipment and all measuring equipment used for this test are adequate to the test and appropriately calibrated.

3.1 Affected Product

- JT8D-217C and JT8D-219 engines
- JT8D-217C and JT8D-219 engines

4 Definition/Abbreviations/Acronyms

Abbreviation / Symbol	Definition / Expansion	Unit
AD	Airworthiness Directive	
ART	Automatic Reserve Thrust	
Band A	Normal Take-Off rating.	
Band A′	Maximum Take-Off rating	
Band B	Maximum Continuous.	
Band C	Maximum Cruise.	
BP	Breather pressure	psi / kPa
CFR	Code of Federal Regulations	
CMM	Component Maintenance Manual	
DAS	Data Acquisition Systems	
EGT	Exhaust Gas Temperature	
EGT_C	Corrected Exhaust Gas Temperature	
EPR	Engine Pressure Ratio (PT7/PT2)	
ESM	Engine Shop Manual	
EUN	Engine Unusual Noise	
F.I.	Flight Idle	
F.P.I.	Fluorescent Penetrant Inspection	
F/N	Thrust	LBS
GUM	Guide to the Expression of Uncertainty in Measurement	
ISO 17025	ISO/IEC 17025	
LPT	Low Pressure Turbine	
M.T.O.	Maximum Take-Off	
MOC	Means of Compliance	
MOP	Main oil pressure	psi / kPa
MOT	Main oil temperature	°F / °C
N.T.O.	Normal Take-Off	
N1	Low Pressure Compressor Rotor Speed	R.P.M
N1 RDT	N1 Run Down Time	sec
N2	High Pressure Compressor Rotor Speed	R.P.M
N2 RDT	N2 Run Down Time	sec
NDT	Non-Destructive Testing	
OBS Fuel Flow	Observe Fuel Flow	LBS/HR
OBS N1	Observe N1	R.P.M
OBS N2	Observe N2	R.P.M

Abbreviation / Symbol	**Definition / Expansion**	**Unit**
OBS Thrust	Observe Thrust	LBS
OC	Oil consumption	gallon/hour, quarts/hour
OEM	Original Equipment Manufacturer	
P/N	Part Number	
PBAR	Barometric Pressure	IN.HGA
PCP	Turbine Cooling Air Pressure	IN.HGA
PLA	Power Lever Angle	
PRBC	Pressure Ratio Bleed Control	
PS2	Compressor Inlet Static Pressure	PSI / KPA
PS3	Low Pressure Compressor Discharge Static Pressure	IN.HGA
PS4	High Pressure Compressor Discharge Static Pressure	IN.HGA
PT2	Compressor Inlet Total Pressure	IN.HGA
PT3	Low Pressure Compressor Discharge Total Pressure	PSI / KPA
PT4	High Pressure Compressor Discharge Total Pressure	PSI / KPA
PT6	High Pressure Turbine Discharge Total Pressure	PSI / KPA
PT7	Turbine Discharge Total Pressure	IN.HGA / PSI / KPA
QC	Quality Control	
RDT	Run-Down Time	
SB	Service Bulletin	
SFC	Specific Fuel Consumption	
TAMB	Ambient Temperature	°F
TSFC	Thrust Specific Fuel Consumption	
TT2 (CIT)	Comp. Inlet Temperature	°F
TT3	Low Pressure Compressor Discharge Total Temperature	°F
TT4	High Pressure Compressor Discharge Total Temperature	°F
TT7 (EGT)	Turbine Discharge Total Temperature	°F
VIB. NO.1	Vibration No. 1 – Inlet (Front) Vibration Pickup	MIL'S
VIB. NO.2	Vibration No. 2 – Exhaust (Rear) Vibration Pickup	MIL'S
VT	Visual Test / Visual Inspection	
WF	Weight of Fuel Flow	LBS/HR

NORMAL TAKE OFF, Band A: The Normal TAKE OFF rating is the maximum thrust normally set for TAKE OFF operation. The rating is set by adjusting the throttle to obtain a predetermined engine pressure ratio. The rating is time limited 5 minutes. This power setting is known as "Band A" by Engine Technical Manuals presented by Figure 3: Power Setting Curve (Figure 1304, Testing–06).

Maximum TAKE OFF, Band A′: The Maximum Takeoff thrust is the maximum thrust certified for takeoff. The rating set by adjusting the throttle to obtain a predetermined engine pressure ratio. In the event of an engine out situation, this rating is provided by the Reserve Takeoff Thrust mechanism when operating at the Normal Takeoff rating. This rating is time-limited to a total of five (5) minutes including the time spent at the Normal Takeoff Rating. This power setting is known as "Band A′ " by Engine Technical Manuals presented by Figure 3: Power Setting Curve (Figure 1304, Testing–06).

Back-to-Back Test: A test comparing performance parameters measured on the same engine in the same facility, before and after replacements of new parts to a test engine. First Test "A" with a base serviceable engine and then "B" test (second) with the same engine but replaced target blades with new ones. So comparing performance parameters on the A and B tests called back-to-back test.

5 Purpose of the Test, Means of Compliance

According to § 33.89 Operation test:

(a) The operation test must include testing found necessary by the Administrator to demonstrate

(1) Starting, idling, acceleration, over-speeding, ignition;

(2) Compliance with the engine response requirements of § 33.73; and

(3) The minimum power or thrust response time to 95 percent rated take-off power or thrust, from power lever positions representative of minimum idle and of minimum flight idle, starting from stabilized idle operation, under the following engine load conditions:

(i) No bleed air and power extraction for aircraft use.

(ii) Maximum allowable bleed air and power extraction for aircraft use.

(iii) An intermediate value for bleed air and power extraction representative of that which might be used as a maximum for aircraft during approach to a landing.

(4) If testing facilities are not available, the determination of power extraction required in paragraph

(a)(3)(ii) and (iii) of this section may be accomplished through appropriate analytical mean.

(b) The operation test must include all testing found necessary by the Administrator to demonstrate that the engine has safe operating characteristics throughout its specified operating envelope. (1)

Also according to § 33.73 Power or thrust response must comply with:

The design and construction of the engine must enable an increase-

(a) From minimum to rated takeoff power or thrust with the maximum bleed air and power extraction to be permitted in an aircraft, without overtemperature, surge, stall, or other detrimental factors occurring to the engine whenever the power control lever is moved from the minimum to the maximum position in not more than 1 second, except that the Administrator may allow additional time increments for different regimes of control operation requiring control scheduling; and

(b) From the fixed minimum flight idle power lever position when provided, or if not provided, from not more than 15 percent of the rated takeoff power or thrust available to 95 percent rated takeoff power or thrust in not over 5 seconds. The 5-second power or thrust response must occur from a stabilized static condition using only the bleed air and accessories loads necessary to run the engine. This takeoff rating is specified by the applicant and need not include thrust augmentation. (1)

5.1 Test Matrix

All tests have been developed in full compliance with the permitted and defined tests specified in the reference document and the Testing section of the Engine Shop Manual (2).

Table 5-1: Certification Tests Matrix

Item	ID	Certification Test / Demonstration	Test Condition	Throttle	Main Parameters	Compliance Criteria	Compliance Evidence
33.89 (a) (1) MOC 4	33.89.a11	Engine Starting	Normal	Starting Initiate normal Monitor complete start sequence	N1 rise EGT rise Oil pressure rise Transient N1/N2 and EGT	Successful start No abnormality EGT limitation for during start Fan response (N1)	-Operator check -Start data plot -Test report
	33.89a12	Engine Starting (Hot day)	Hot Day Ambient condition	Perform engine start at specified hot-day condition	N1 rise EGT rise Oil pressure rise Transient N1/N2 and EGT	Successful start No abnormality EGT limitation for during start Fan response (N1)	-Operator check -Test report
	33.89a13	Re-Starting	Hot Start / Restart	Engine restart	N1 rise EGT rise Oil pressure rise Transient N1/N2 and EGT	Successful start No hung start EGT limitation for during start Fan response (N1)	-Test report
	33.89a20	Idle Operation Test	-	Stabilize at Ground idle	Time to idle N2 EGT limitation for stable ground idle Vibes	Stable idle operation Core response (N2) Repeatability	-Stabilized idle data -Test Report
	33.89a21	Flight Idle Stability	-	Stabilize at Flight idle	N1 Rise Vibrations limit	Stable operation Repeatability	-Stabilized idle data
	33.89a30	Acceleration (Slow and Normal)	Idle to NTO	Acceleration & Deceleration	N1 Thrust EGT Vibes	Fan to Throttle Response EGT Response Thrust Response limitation No Surge/Stall or over-speed Reputability	-Transient plots -Test report
	33.89a31	Over-speed Protection Test	Hot Day Ambient	NTO, MTO & ART	N1 Thrust Vibes	EGT limitation Over-speed protection check Fuel control response No sign of over-speed, over-temp., surge or stall	-Protection-function evidence
33.89 (a) (2)	-	MOC 0			-	-	

Item	ID	Certification Test / Demonstration	Test Condition	Throttle	Main Parameters	Compliance Criteria	Compliance Evidence
33.73 (a) MOC 4,2	33.73a10	Rapid Accel. Check (Data to verify engine model)	No bleed No extraction F.I (Max Bleed +Max power in model)	Power lever snap to MAX in ≤1 s	Thrust N2 N1 EGT PLA Time	Rated takeoff thrust reached without surge/stall/over-temp. Response to snap lever movement demonstrated	-Test Report -Compliance Statement
33.73 (b) MOC 4	33.73a11	5-sec Engine Response	Engine/accessory loads only Flight idle	Move power lever to 95% rated takeoff in 5 sec.	Time to 95% thrust Thrust N2	Time to 95% thrust ≤5 s Stable operation	-Thrust-vs-time plot -Test Run Sheet
33.89 (a) (3) (i) MOC 24	33.89a30	Accel. Response GI– No Aircraft Loads	No Bleed No Power Extraction Stabilized G.I	Accelerate to 95% Rated Takeoff Thrust	Transient Data including N2 Thrust EGT PLA Position Time 95%	Determine minimum response time Absence of unacceptable surge/stall over-speed, over-temp or vibration.	-Response-time calculation -Test Run Sheet
	33.89a31	Accel. Response FI– No Aircraft Loads	No Bleed No Power Extraction Stabilized Flight Idle	Accelerate to 95% Rated Takeoff Thrust	Transient Data including N2 Thrust EGT PLA Position Time 95%	Determine minimum response time Repeatability Absence of unacceptable surge/stall over-speed, over-temp or vibration.	-Response-time Calculation -Test Run Sheet
33.89 (a) (3) (ii)	-	MOC 2,4 Acceleration – Maximum Bleed & Extraction	Stabilized Ground Idle; maximum allowable bleed + maximum aircraft power extraction	-	-	-	Test data / analytical substantiation
	-	Acceleration – Maximum Bleed & Extraction / Flight Idle	Stabilized Flight Idle; maximum allowable bleed + maximum aircraft power extraction	-	-	-	Test data / analytical substantiation
33.89 (a) (3) (iii)	-	MOC 2,4 Acceleration – Approach Bleed/Extraction	Stabilized Ground Idle; representative approach bleed and power extraction	-	-	-	Test data / analytical substantiation
	-	Acceleration – Approach Bleed/Extraction / Flight Idle	Stabilized Flight Idle; representative approach bleed and power extraction	-	-	-	Test data / analytical substantiation
33.89 (a) (4)	-	MOC 0	-	-	-	-	-
33.89 (b) MO	33.89b10	Functional	Ground Idle, Part Power, MCR NTO and applicable operating points	According to ESM	Vibes. Thrust N2 EPR Ambient Temperature	Vibration Survey Part Power Trim check MCR & NTO Check	-Test Run Sheet -Vibration survey report

Item	ID	Certification Test / Demonstration	Test Condition	Throttle	Main Parameters	Compliance Criteria	Compliance Evidence
C 2 3 4	33.89b11	Transient Envelope Test	G.I, F.I Band A, B, C	Move between stable points	Trust N2 EGT	Temperature Response, No detrimental engine behavior	-Transient Graphs
	33.89b12	Vibration Survey	Idle MTO	Acceleration Deceleration	Vibration amplitude N2 N1	Within approved limits	-Vibration survey report
	33.89b12	Surge/Stall Demonstration Rapid Accel. Check	-	Perform rapid acceleration checks	Thrust EPR, N2 N1 EGT	No surge/stall	-No Sign of Abnormality
	33.89b20	Performance-Operating Envelope Test	Ground Idle Flight Idle Part Power Max. Cr. (Band C) Max. Cont. (Band B) NTO (Band A) MTO (Band A') ART	Stabilize and operate engine at specified points	Thrust N2 EGT EPR	Safe operating characteristics	-Perfor. Test Report -Test Run Sheet -Safety Assessment
	33.89b30	Repeatability Tests	Different Ambient Condition		Thrust N2 EPR Ambient Temperature	Consistent response	-Repeat. analysis

5.2 Test procedure

For the engine test program of the manufactured LPT blade third and fourth stages, the following procedure is defined:

Initially, an engine with OEM parts is tested in accordance with the JT8D-219 engine manual's acceptance procedure. The performance results of this engine recorded as the baseline. Subsequently, the engine should disassemble, and the OEM 4th-stage blades would replace with the manufactured blades, with the test results used for comparison. Upon successful completion, this process will be repeated for the LPT blade third stage, and a test will be conducted with both the 3rd and 4th-stage manufactured blades installed.

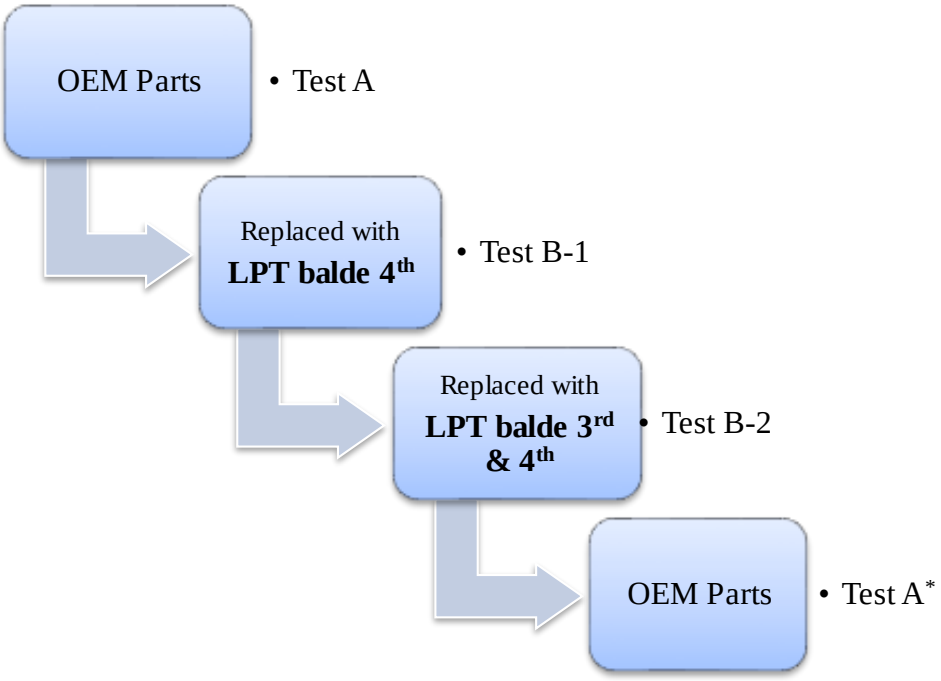

Figure 5-1: Back-to-back test procedure

First Test "A" with a base serviceable engine and then "B" test (second) with the same engine but replaced target blades with new ones. So comparing performance parameters on the A and B tests called back-to-back test. Therefore, comparative tests with emphasis on LPT thermal and structural stresses shall be accomplished to ensure turbine, and engine, safe operation and sustained performance. Comparative (back-to-back) tests designed to verify performance differentials of the turbine and engine, primarily for testing before and after blade replacements.

This test would be accomplish by using an engine with known stable performance that is capable of operating over the engine's entire thrust range. A serviceable JT8D-219 engine that has successfully passed the acceptance test with OEM parts must be use. Performance parameters must be within the acceptable range, and the testing procedure must be conduct in accordance with the ESM instructions. Before blades replacement in the engine that may effect on performance, the engine must be run through the first stage of a back-to-back test. The schedule includes all thrust rating points that would be recorded during routine acceptance testing and include other data points.

The data from the first run ("A") should be carefully examining to determine if engine performance is stable. Then fourth stage LPT blade replacement should accomplish on the same engine and be test in the test-cell with the same schedule as initially tested.

Before the first test, the blade tip clearance and notch torque must be measure and record. Additionally, initial hand rotating and dry cranking of the engine must be perform without any issues.

To ensure safety during testing, the procedure of test B is carry out in several stages. At the end of each stage, the engine would shut down for inspections; if no evidence of abnormalities are observe in either the inspections or the recorded parameters, the next test stage will proceed. The final stage is a performance test to record parameters for comparison with Test A (the engine with OEM parts). Therefore, the following tests and inspections are required:

Table 5-2: Precautionary Check and Tests

	Sequence	Rated Thrust
Step 1	IDLE stabilization check	Ground Idle- Flight Idle
Step 2	Part Power stabilization check	Part Power
Step 3	Take-off check / Vibration Survey	Take-off
Step 4	Acceptance/Performance Test	Idle, Part Power, Take-off, ART

1. **IDLE stabilization:**

At the first scheduled IDLE stabilization, the engine shall be shut down via the normal procedure to perform the required inspections of the Stage 4 turbine blades. Oil pressure and temperature values, N1, N2, run-down time, and time to idle must be record and evaluated; furthermore, the engine should undergo a visual inspection for any signs of leakage. The engine exhaust must also be inspect for any evidence of metallic debris. Additional inspection items may be included based on the engine condition and the discretion of Quality Control (QC) staff.

2. **Part Power stabilization:**

At the first scheduled Part Power stabilization, a normal shutdown shall be perform followed by a re-inspection of the Stage 4 turbine blades. Any further checks deemed necessary by QC based on the engine's status shall also be conducted at this stage (like borescope check, Torque notch measurement and …).

3. **IDLE / Take-off sequence:**

The engine shall be re-started and stabilized at IDLE for 5 minutes, then advanced the throttle slowly (during 5 minutes) to Take-Off power and stabilized for 3 minutes. After a normal shutdown, the Stage 4 turbine blades shall be inspected. N1, N2, vibrations, EGT, thrust and EPR values must be recorded, and the following inspections should be performed: engine inlet check for oil leaks, exhaust area check, and magnetic chip detector inspection. Additionally, for verification purposes, the notch torque inspection must be perform again. Additional inspections may be perform depending on the engine condition and QC requirements (like borescope check, Torque notch measurement and …).

NOTE: Vibration, N1, N2 and EGT parameters should be carefully monitored during throttle movement from idle to take-off power.

4. Functional and performance Test:

Following above steps, the engine will be re-started to complete the remaining test schedule as a full test run according to acceptance procedure of the JT8D-219 ESM and record performance parameters (2).

Functional and performance test procedure is shown in Figure 5-2. This test procedure should be perform on each configuration engines. "A" test with serviceable engine and "B-1" and "B-2" test with new bladed engine. In this common way to evaluate the new blade performance, by checking the differences in performance parameters with and without installing new blades.

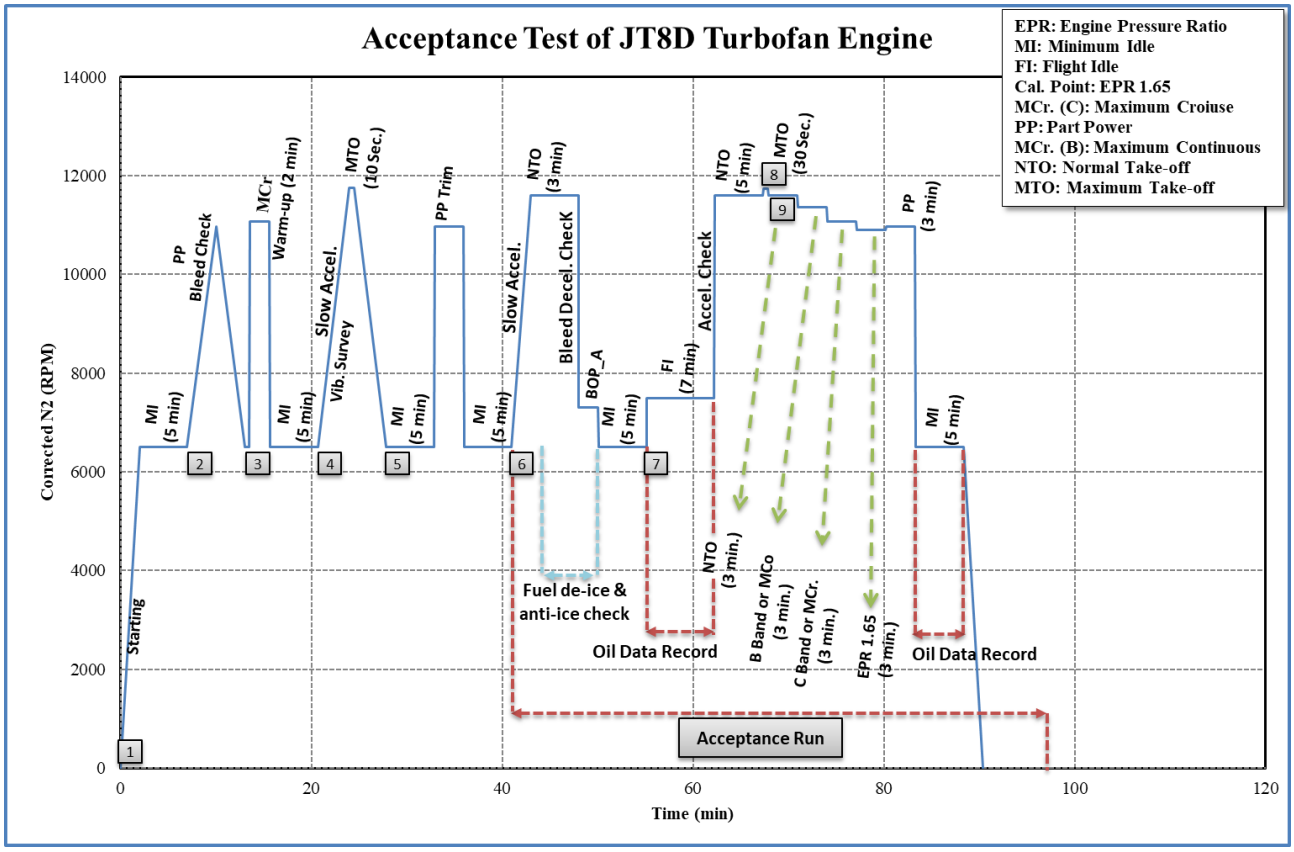

Figure 5-2: Procedure of step 4 of Test B/ Acceptance test

Data from these runs should be analyzed to determine the effects of new blades. The engine performance parameters must not only fall within the acceptance limits specified in the ESM, but they should also differ only slightly from Test A (the test with OEM parts). The acceptance limits are provided in the tables below. Test program shown in Table 5-3.

Table 5-3: Test Program

ID	Test / Requirement	To be performed at test:													
		Ground Idle	Flight Idle	Bleed Valve Check	MCR (Band C)	Vibration Survey	P.P Trim check	NTO (Band A)	Accel. Check	MTO (Band A')	MCO (Band B)	MCR (Band C)	Part Power	Cal. Point	5-sec Engine Response
33.89a11	Engine Starting	1				2									
33.89a12	Engine Starting (Hot Day)	1				2									
33.89a13	Re-Starting	1				2									
33.89a20	Idle Operation Test	1	1			2									
33.89a21	Flight Idle Stability		1			2									
33.89a30	Acceleration (Slow and Normal)	1	1			1		3	1						
33.89a31	Over-speed Protection Test			1		2			2				2		
33.73a10	Rapid Accel. Check (Data to verify engine model)					3		3	1	3					
33.73a11	5-sec Engine Response								2						1
33.89a32	Accel. Response GI – No Aircraft Loads	1				2		3	1						3
33.89a33	Accel. Response FI – No Aircraft Loads		1			2		3	1						3
33.89a34	Accel. Response GI – Maximum Bleed & Extraction	1		2		2		3	1						3
33.89a35	Accel. Response FI – Maximum Bleed & Extraction		1	2		2		3	1						3
33.89a36	Accel. Response GI – Approach Bleed/Extraction	1		2		2		3	1						3
33.89a37	Accel. Response FI – Approach Bleed/Extraction		1	2		2		3	1						3
33.89b10	Functional Test	1	1		1	2	2	3		1	1	1	1		
33.89b11	Transient Envelope Test	1	1		1	2		3	1	1	1	1	1		
33.89b12	Vibration Survey	1	1				3	2		1	3	3	2		
33.89b13	Surge/Stall Demonstration – Rapid Accel. Check			2		3		3	1	2					
33.89b20	Performance / Operating Envelope Test	1	1		1	2	2	1		1	1	1	1	1	
33.89b30	Repeatability Tests	1	1		1	3		1		3	3	3	3	3	

Priority legend: 1 = purpose / test point; 2 = observed during test; 3 = recorded during test and/or limited observation.

5.3 Test Instructions

Before initial start:

(1) , (2) Fuel pump filter pressure drop switch circuit test

(3) Start engine

(4) Visual inspection of entire engine for any problems such as fuel or oil leaks

(5) Adjust engine main oil pressure

Table 5-4: Engine Start

Test:	1. Starting		Purpose:	Functional check & Part 33-89 (a) 1	
Step	N2 (rpm)/PLA	Duration	Record/Monitor	Check	Criteria
C.(1).(g)	2450		-		EGT rise
C.(1).(h)	4000		EGT (max)	Oil pressure	Limits of starting
			Fuel Flow (max)		No flame at ext.
			Start time[1] (sec)		No rapid EGT raise
					Light up in 20s
I.(1)	IDLE	2-3 min		Oil temperature	100°F (38°C) (min)
S.(6)	IDLE		Ice Formation[2]		

(7) Normal compressor bleed valve operation check, Table 5-5**.**

(a) Slowly accelerate the engine from IDLE and check bleed closing point. Record PS3 and PT2 at the bleed closing point. Bleed closure is indicated by a sudden increase in the valve supply pressure (PS4) from ambient.

(b) Slowly decelerate the engine from approximately 500 pounds (227 kg) thrust above the bleed closing point (BOP[3]_A) and check bleed opening point. Record PS3 and PT2 at the bleed opening point. Bleed opening is indicated by a sudden decrease in the valve supply pressure (PS4).

(c) and (d) Bleed valve events should occur between 8500 and 9000 rpm N2.

(e) If no EPR change is noted, a bleed valve operation fault is assumed.

[1] Time required from energizing of starter to idling speed

[2] Check for icing. If icing conditions are indicated, testing should be discontinued.

[3] Bleed Opening Point

Table 5-5: Normal compressor bleed valve operation check

Test:	**2. bleed valve operation**		Purpose:	Functional check & Part 33-89 (a) 1	
Step	N2 (rpm)/PLA	Duration	Record/Monitor	Check	Criteria
S.(7).(a)	IDLE to BCP[1]	Slow Accel.	PS3	PS4 (rise)	Figure 1307, Testing −06
	8500-9000		PT2		
S.(7).(b)	BOP_A to IDLE	Slow Decel.	PS3	PS4 (drop)	Figure 1307, Testing −06
	8500-9000		PT2		

(8) Warm up engine at MAXIMUM CRUISE thrust for two minutes.

(9) Run five minutes at IDLE and record oil level and OIL−IN temperature, Table 5-6.

Table 5-6: Engine Warm-up

Test:	**3. Warm up**		Purpose:	Functional check & Part 33-89 (a) 1	
Step	N2 (rpm)/PLA	Duration	Record/Monitor	Check	Criteria
S.(7).(a)	MAX CRUISE	2 Minutes	Oil level	-	-
	IDLE	5 Minuets	T OIL−IN	-	-

(10) Perform engine vibration survey, Table 5-7.

(1) An engine vibration survey shall be conducted on a Warm Engine **(8)**. If the engine is shut down, this Warm Up must be repeated before conducting a vibration survey.

(2) The vibration survey shall consist of the following:

(a) Stabilize 5 minutes at DESCENT/GROUND IDLE. Energize the Automatic Reserve Take−Off Thrust (ART) solenoid. The ART system check is acceptable if the ART indicator light is energized.

(b) Make a slow (two to three minute) acceleration from idle speed to MAXIMUM TAKE−OFF thrust (Band A' of Figure 3). Record peak observed inlet and rear case vibration amplitudes and N1 and N2 rpm at which they occur.

(c) Stabilize 30 seconds at MAXIMUM TAKE−OFF thrust (Band A' of Figure 3).

(d) Continue vibration survey by a slow (two to three minute) deceleration to IDLE. De−energize ART solenoid. Record peak observed inlet and rear case vibration amplitudes and N1 and N2 rpm at which they occur. If vibration limits are exceeded during transients, steady state vibration shall be recorded at approximately the speed which produced peak vibration to determine if vibration is acceptable per limits of Table 6-2.

(3) The equipment necessary to measure vibration shall fall within the conditions set in "Vibration System Response Characteristics", Figure 1301, Testing−02. Such a system will include meter, filter (40 cps), and pickups.

[1] Bleed Closing Point

Table 5-7: Vibration survey

Test:	**4. Vibration survey**		Purpose:	Functional check & Part 33-89 (a) 1	
Step	N2 (rpm)/PLA	Duration	Record/Monitor	Check	Criteria
K.(2).(a)	IDLE	5 Minutes		ART System	
K.(2).(b)	IDLE to M.T.O.	2-3 Minutes	Peak Vibrations		Table 6-2
			N1 at Peak Vib.		
			N2 at Peak Vib.		
K.(2).(c)	M.T.O.	30 Seconds	Peak Vibrations		Table 6-2
K.(2).(d)	M.T.O. to IDLE	2-3 Minutes	Peak Vibrations		Table 6-2
			N1 at Peak Vib.		
			N2 at Peak Vib.		

(11) With fuel control at IDLE power lever position (38.4 degrees at engine cross shaft) adjust fuel control idle trim adjustment within N2 speed band for inlet temperature at the time as shown in Figure 4.

(12) Part Power Trim Setting

(a) With power lever against Standard (S) part power trim stop, Figure 1308, Testing−06, adjust fuel control Max. trim adjustment so that PT7/PT2 falls within acceptable limits shown in Figure 5. For conditions of TT2 equal to 90°F (32°C) or less, detach the part power stop, index it to Cold (C) position, reinstall and trim to Cold Setting Band, Figure 5.

(13) Remove the remote trimmers from the fuel control. Stabilize for three minutes with the power lever against the part power stop and record Thrust, N2, TT2, PT2, and PT7. Check PT7/PT2 against the requirements of Figure 5. If this value is not within the acceptable limits, trim the control until an acceptable final trim check is made with the trimmers removed. After an acceptable final trim to requirements of Figure 5 has been obtained, wire the part power trim stop in the stowed position.

Note: If possible, to increase confidence in the comparative analysis of Tests A and B, the Part Power Trim process is to be omitted. Additionally, it is imperative to confirm that the engine parameters are maintained within the specified tolerance ranges.

Table 5-8: Power Trim

Test:	**5. Power Trim**		Purpose:		
Step	N2 (rpm)/PLA	Duration	Record/Monitor	Check	Criteria
S.(11)	IDLE (38.4° exact)	Not Specified	N2		Figure 4
S.(12)	Part Power	Not Specified	PT7/PT2		Figure 5
S.(13)	Part Power	3 minutes	Thrust	PT7/PT2	Figure 5
			N2		
			TT2		
			PT2		
			PT7		

(14) Visually inspect the entire engine for any problems such as fuel or oil leaks at this time (The Engine Check Chart). Any leaks shall be repaired before continuance of engine test. If oil leaks are encountered, or if repairs are made to the engine which results in a loss of oil from the system, additional oil shall be added.

(15) Perform "<u>Acceptance Test</u>" run according to schedule which follows:

Test:	Acceptance Test		Purpose:		
Step	N2 (rpm)/PLA	Duration	Record/Monitor	Check	Criteria
S.(15) Through S.(27)	-	-	-		

(16) Warm up by a slow acceleration, in increments, to NORMAL TAKE−OFF thrust.

(17) Run three minutes at NORMAL TAKE−OFF, Band A, Figure 3. Record data.

(a) Check operation of fuel de−icing system (for engines so equipped) during this NORMAL TAKE−OFF running. Open de−icing air valve and observe change in fuel temperature downstream of heat exchanger. Fuel temperature must increase a minimum of 75°F (42°C) in less than one minute after valve is opened. Do not adjust power lever for resultant loss of thrust. Do not allow fuel temperature to exceed 200°F (93°C).

(b) During NORMAL TAKE−OFF running, actuate anti−icing system and record readings to indicate satisfactory operation of anti−icing system. It is not necessary to record readings of engine performance when anti−icing system is in operation.

(c) Deceleration Bleed Control System Check. Decelerate to the power lever position that is just above the bleeds opening position (approximately 500 pounds thrust above opening point BOP_A) by a power lever movement in not more than one second. The system is operating normally if the bleed supply pressure (PS4) drops to near ambient soon after the throttle movement and then increases to approximately PS4 level as the engine stabilizes. Decelerate to ground/descent idle and stabilize 5 minutes.

(d) Run five minutes at IDLE speed. Record data including oil−in temperature and oil level for oil consumption evaluation.

(e) With the fuel control power lever at the idle position, de−energize the idle select solenoid and record data. N2 speed increase should be within the limits.

Table 5-9: Systems Operation Check

Test:	6. Systems Operation Check		Purpose:		
Step	N2 (rpm)/PLA	Duration	Record/Monitor	Check	Criteria
S.(17)	N.T.O.	3 minuets	Record data		
S.(17).(a)				Fuel Temp (rise)	75°F (42°C)
					200°F (93°C)
S.(7).(c)	N.T.O. to BOP_A	Snap [1] Decel.	-	PS4 (drop)	Near ambient

[1] Less than 1 second

Test:	6. Systems Operation Check		Purpose:		
Step	N2 (rpm)/PLA	Duration	Record/Monitor	Check	Criteria
				PS4 (rise)	
S.(7).(d)	IDLE	5 minuets	Record data		
			T Oil-in		
			Oil level		
S.(7).(d)	IDLE			N2 Increment	

(18) Engine Acceleration Check

(1) Determine 99% of Normal Take−Off N2. If Normal Take−Off data has not been recorded within the current sequence of engine operation, perform the following. Advance power lever to Normal Take−Off thrust position (30 second power lever transient). Stabilize for 60 seconds and record data.

(2) Decelerate to DESCENT/GROUND IDLE position. De−energize idle select solenoid and stabilize for seven minutes. Record data point.

(3) Acceleration check

(a) Set the power lever to Normal Takeoff (Band A of Figure 3) in no more than 1.0 second and let the engine become stable for 60 seconds.

(b) Pull the power lever back to Approach Idle.

(c) Record the expired time from when the power lever movement started to when the engine gets to the 99 percent of Normal Takeoff N2 rpm recorded in step (1).

(d) When the acceleration time check is completed, immediately pull the power lever back to Approach Idle.

(4) Acceleration time shall not exceed limits in Figure 6.

Table 5-10: Acceleration Check

Test:	7. Acceleration Check		Purpose:		
Step	N2 (rpm)/PLA	Duration	Record/Monitor	Check	Criteria
S.(18) Ref.					
J.(3).(a)	IDLE to N.T.O	Snap Accel.			
	N.T.O	60 seconds			
J.(3).(c)- J.(4)	A. IDLE to N.T.O	20-25 seconds			
			Time N2 Reach 0.99 N.T.O (1)		Figure 6

(19) After you complete the acceleration check and while the engine is at Normal Take−Off power let the engine become stable for five minutes before you do a check of the Automatic Reserve Thrust (ART) operation.

(a) Energize the automatic reserve thrust (ART) solenoid and stabilize for a maximum of 30 seconds. Record data and save for final log. The engine is acceptable if the time to achieve the N2 speed increase of ESM, does not exceed 1.5 seconds from solenoid

actuation and the automatic reserve thrust (ART) increment is within the limits of ESM. If the automatic reserve thrust (ART) increment is more than the maximum limits of ESM. see "Paragraph 1.S.(27) fuel control adjustment procedure" of Engine Shop Manual.

(b) Following data acquisition, de−energize automatic reserve take−off thrust (ART) solenoid.

Table 5-11: ART Check

Test:	8. ART Check		Purpose:		
Step	N2 (rpm)/PLA	Duration	Record/Monitor	Check	Criteria
S.(19)	N.T.O	5 minuets			
S.(19).(a)	ART	30 seconds	Record data	N2 Increment	
			Final log		

(20) Run a minimum of three minutes for stabilizing power at Normal Take−Off, Band A, Figure 3; then record data.

(a) Record average and individual EGT readings from each exhaust gas system (TT7) probe. Compare the corrected average EGT (TT7/theta T2, °K) to Figure 7to determine EGT margin at Normal Takeoff Rating. The maximum−to−minimum spread shall not exceed 200°F (111°C).

Table 5-12: N.T.O Performance Check

Test:	9. N.T.O Performance Check		Purpose:		
Step	N2 (rpm)/PLA	Duration	Record/Monitor	Check	Criteria
S.(20)	N.T.O	3 minuets	Record data		
S.(20).(a)			Average EGT	EGT margin	Figure 7
			All TT7s		Δ<200°F (111°C)

(21) Run a minimum of three minutes for stabilizing power at calibration Band B, Figure 3; then record data.

Table 5-13: Calibration Power Performance Check

Test:	9. Calib. Power Performance Check		Purpose:		
Step	N2 (rpm)/PLA	Duration	Record/Monitor	Check	Criteria
S.(21)	Calib. Power	3 minuets	Record data		

(22) Run a minimum of three minutes for stabilizing power at maximum cruise, Band C, Figure 3; then record data.

Table 5-14: Maximum Cruise Performance Check

Test:	9. Max. Cruise Performance Check		Purpose:		
Step	N2 (rpm)/PLA	Duration	Record/Monitor	Check	Criteria
S.(22)	Max. Cruise	3 minuets	Record data		

(23) Run a minimum of three minutes for stabilizing power at a calibration point of 1.65 EPR (14,700 − 15,200 pounds thrust (6668 − 6895 kg); then record data.

Table 5-15: Calib. Point 1.65 EPR Performance Check

Test:	9. 1.65 EPR Performance Check		Purpose:		
Step	N2 (rpm)/PLA	Duration	Record/Monitor	Check	Criteria
S.(23)	1.65 EPR	3 minuets	Record data		
S.(24) optional				TT7s	Δ<200°F (111°C)

(24) If desired, make a part power exhaust gas temperature spread check. Part power temperature spread shall not exceed 200°F (111°C).

(25) , (26) Retard power lever to IDLE position and operate engine for five minutes. Stabilize until oil–in temperature coincides with the initial idle stabilized temperature at start of acceptance. Record oil level and oil–in and oil–out temperature at this point. Determine oil consumption in gallons per hour using this figure and the figure recorded at beginning of this test.

Table 5-16: Final Oil Consumption Check record

Test:	Oil Consumption Check		Purpose:		
Step	N2 (rpm)/PLA	Duration	Record/Monitor	Check	Criteria
S.(25)	IDLE	5 minuets	Oil level		
			T Oil-in		
			T Oil-out		
			Oil Consumption		

(27) Shut down engine. Perform visual inspection at this time of entire engine for any discrepancies such as fuel or oil leaks.

(1) Power lever – IDLE

NOTE. Before shutting down from power settings above maximum cruise power, idle the engine for five minutes to allow uniform cooling; however, if an emergency stop is made from such settings without idling, check both rotors for freedom of rotation before the next start.

(28) Check for static oil leakage through No. 6 bearing carbon bon seals, to be measured for a 20 minute period initially following engine shutdown. Oil leakage appearing as an oily wetness but not puddling during this 20 minute period is acceptable. Oil puddling which occurs before the expiration of 20 minutes is not acceptable.

(29) Remove fuel control inlet strainer and inspect it. Note results of the inspection in engine log. Clean and install uncontaminated fuel control inlet strainers and perform a leak check.

(30) Remove fuel pump interstage filter and inspect it. The old filter must be torn apart carefully and thoroughly inspected between each pleat. In addition, inspect fuel pump interstage filter housing cover. Note results of inspection in log. Install a new fuel pump interstage filter and perform a leak check.

(31) Upon completion of test, check ignition system by listening that both igniter plugs are firing with the start system energized and that the right igniter fires normally when the AC continuous system is energized. For engines having a left and right exciter, check by listening that the left igniter is firing when the left exciter is energized and the right igniter is firing when the right exciter is energized.

6 Proposed Test Results (Fail/Pass Criteria)

The engine should meet all specified performance criteria, which can be checked by the acceptance test procedure. At the end of the tests, the test engine and its components should operate satisfactorily and no part failures should be observed during inspections. Engine temperature (EGT), engine thrust, EPR and SFC shall be determined at the same rated thrust values during both A and B (back-to-back) tests. The comparative tests shall be considered satisfactorily completed when:

a. During the B-1 and B-2 tests, engine performance and operation shall be within established limits than that of a nature and rate specified in the engine specification (See Test procedure).

b. The component teardown inspection shows no indication of failed, excessively worn, distorted, or weakened parts. During this inspection, the blades shall be completely disassembled for examination, and measured as necessary to disclose excessively worn, distorted, and NDT checks.

c. These parameters shall be compare with values for the base serviceable engine (test A) and new bladed engine tested (Test B-1 and B-2) with similar measurements:

- EPR
- Thrust
- EPR/Thrust relation
- EGT limit
- N1 limit
- N2 limit
- SFC limit
- Vibration No.1 limit
- Vibration No.2 limit

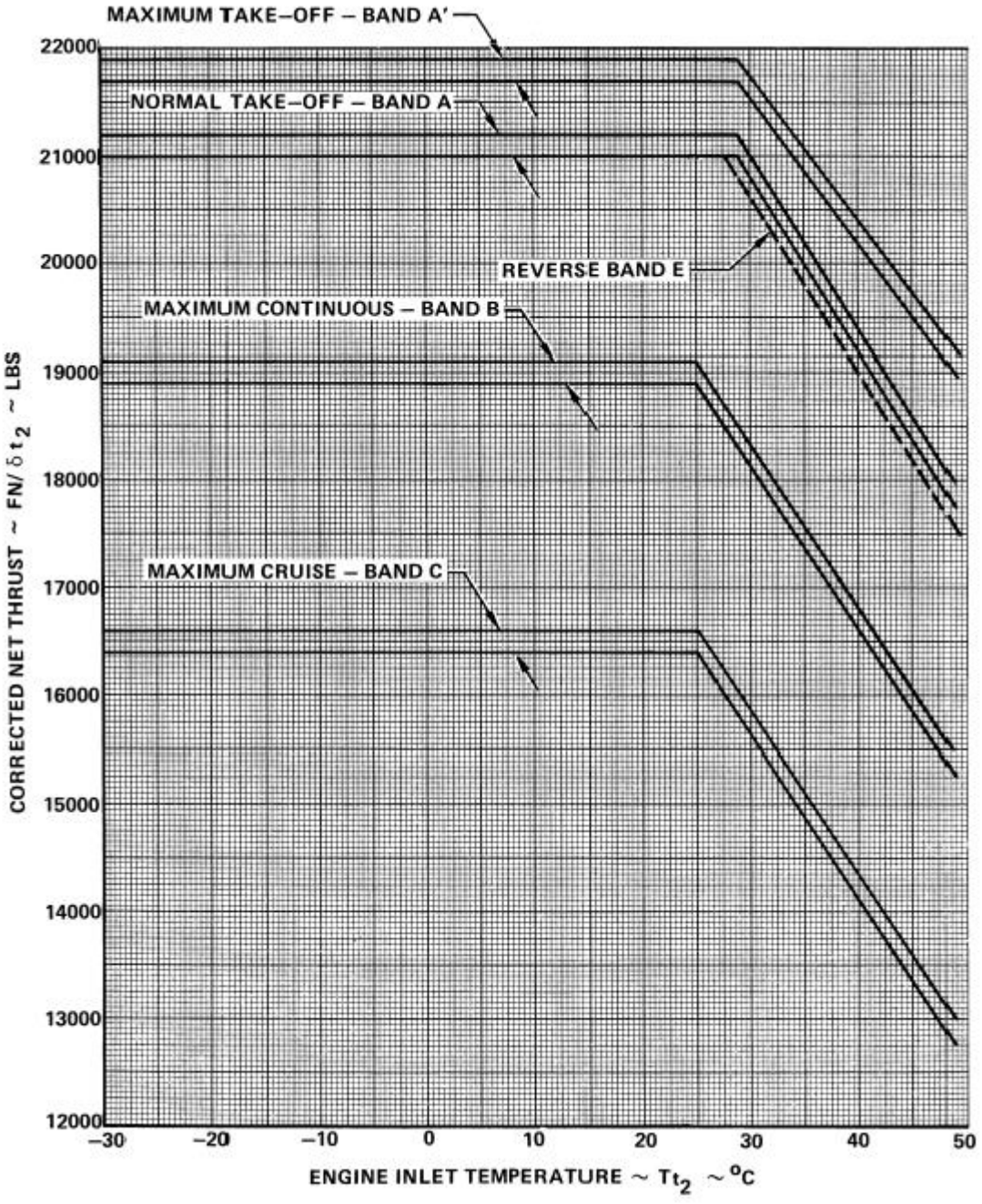

Figure 3: Power Setting Curve (Figure 1304, Testing–06)

Table 6-1: Maximum Safe Operating Vibration Limits (1302 in Testing–06)

Pickup Location	Single Amplitude	Double Amplitude
INLET SECTION		
Overall	0.004 In. (0.102 mm) (4.0 mils)	0.008 In. (0.203 mm) (8.0 mils)
Low Rotor Frequency	0.0035 In. (0.089 mm) (3.5 mils)	0.007 In. (0.178 mm) (7.0 mils)
High Rotor Frequency	0.0025 In. (0.064 mm) (2.5 mils)	0.005 In. (0.127 mm) (5.0 mils)
REAR SECTION		
Overall	0.003 In. (0.076 mm) (3.0 mils)	0.006 In. (0.152 mm) (6.0 mils)

Pickup Location	Single Amplitude	Double Amplitude
Low Rotor Frequency	0.003 In. (0.076 mm) (3.0 mils)	0.006 In. (0.152 mm) (6.0 mils)
High Rotor Frequency	0.002 Inch (0.051 mm) (2.0 mils)	0.004 In. (0.102 mm) (4.0 mils)

Table 6-2: Acceptance Limits Vibration Amplitudes (1303 in Testing–06)

Pickup Location	Single Amplitude	Double Amplitude
INLET SECTION 0.003 In.	0.0015 In. (0.038 mm) (1.5 mils)	0.003 In. (0.076 mm) (3.0 mils)
REAR SECTION	0.0015 In. (0.038 mm) (1.5 mils)	0.003 In. (0.076 mm) (3.0 mils)

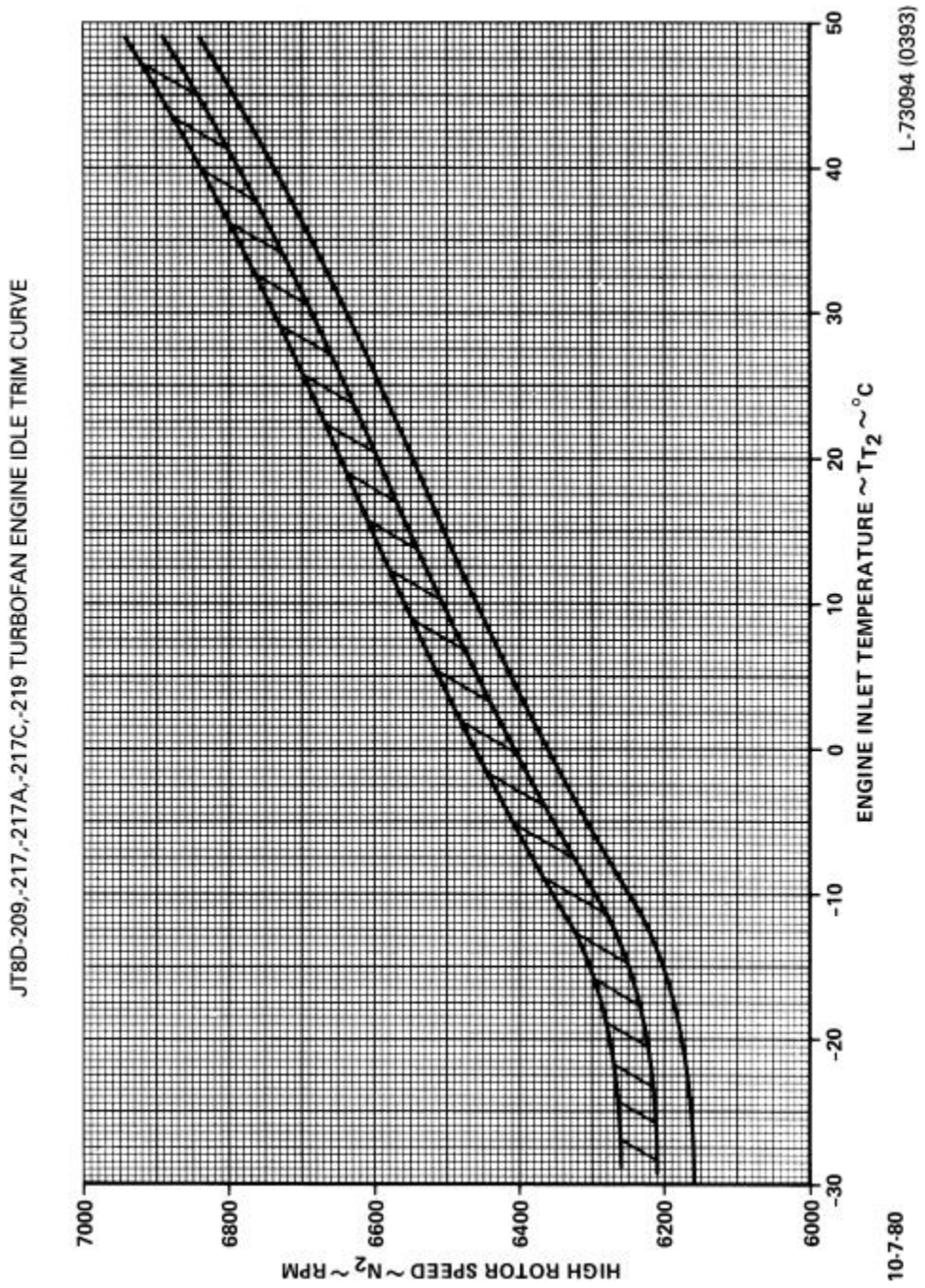

Figure 4: Idle Trim Curve (Figure 1305, Testing–06)

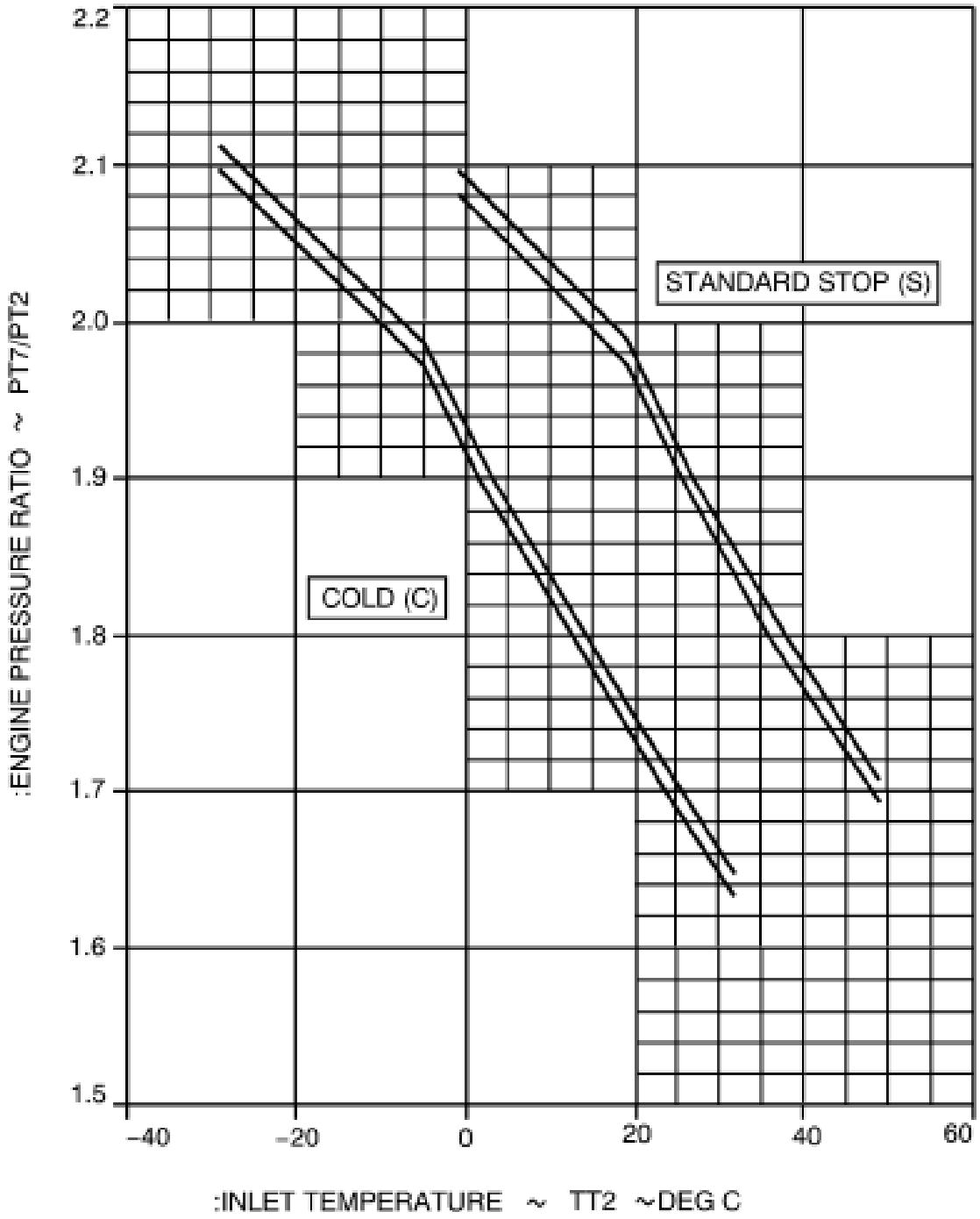

Figure 5: Part Power Trim Curve (Figure 1309, Testing–06)

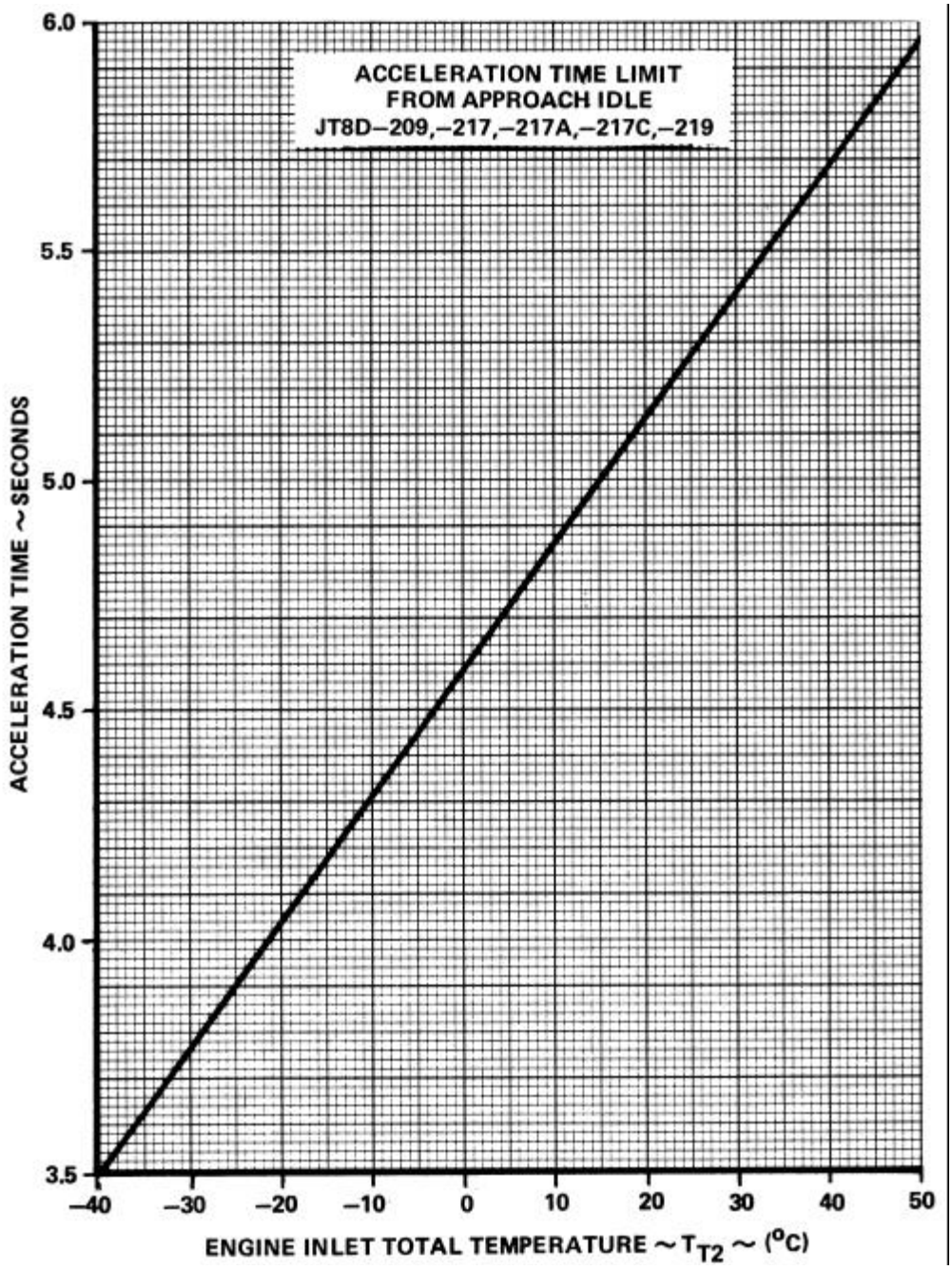

Figure 6: Acceleration Time Limit From Approach Idle (Figure 1306, Testing–06)

NOTE:

1. CORRECTED EGT= $\dfrac{(T_{T7}\ \text{OBSERVED}\ ^{o}C + 273)}{\left(\dfrac{T_{T2}\ ^{o}C + 273}{288}\right)^{1.019}}$ / 1.019

2. EGT MARGIN IS DEFINED AS THE DIFFERENCE BETWEEN THE ENGINE $T_{T7}/\theta\ T2^{1.019}$ AND THE LINE AT CONSTANT EPR

3. THE EGT LIMIT REPRESENTS AN ENGINE WITH ZERO MARGIN TO THE NORMAL TAKE-OFF LIMIT (ORANGE LINE) ON A 29 °C AMBIENT DAY.

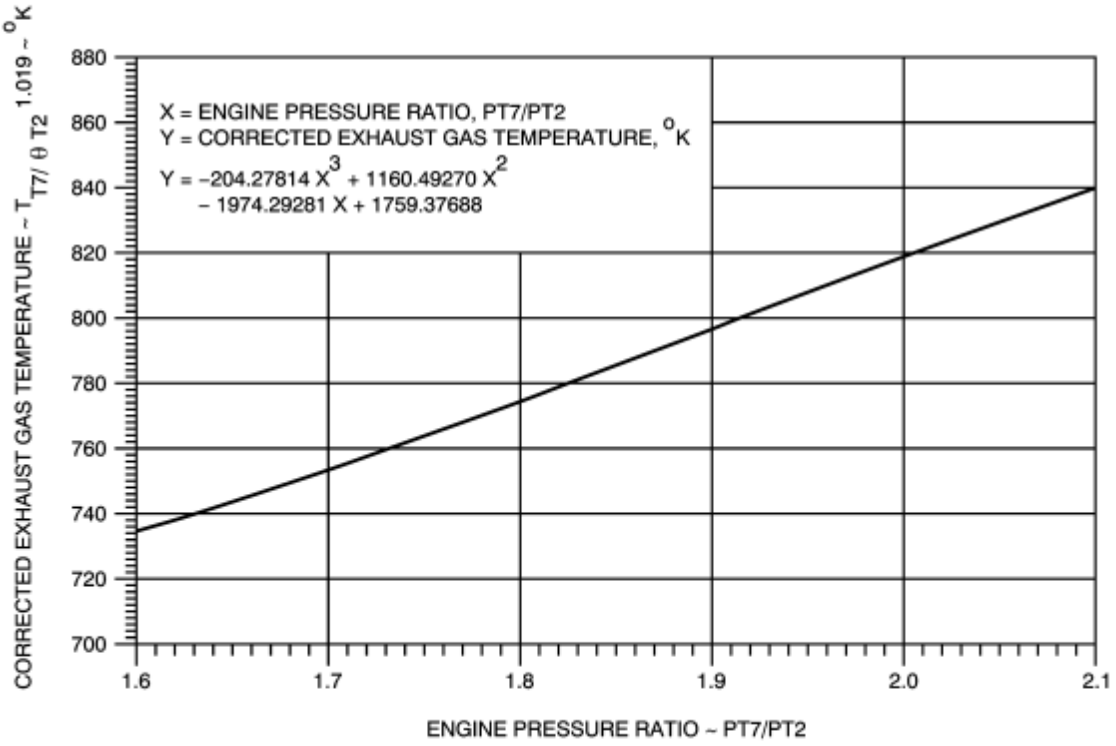

Figure 7: EGT Margin Check Curve (Figure 1313, Testing–06)

Table 6-3: Engine Check Chart For JT8D–219 (Table 1304 in Testing–06)

Power Setting	Corrected Jet Thrust LBS (kg)	Corrected Low Pressure Compressor RPM (N1)	Corrected High Pressure Compressor RPM (N2)	Corrected TSFC LBS/HR/LB (kg/HR/Kg) Thrust
MAXIMUM TAKE-OFF	21,700 (9,841)	7,860	11,760	–
NORMAL TAKE-OFF	21,000 (9,524)	7,670	11,680	0.515 (0.515)
Maximum Climb	18,900 (8,573)	7,240	11,440	0.489 (0.489)
Max. Cruise	16,400 (7.439)	6,890	11,170	0.481 (0.481)

7 Specification of the Test Specimen

The JT8D-200 series engine is an axial flow, dual spool, medium bypass ratio, fully ducted turbofan.

The JT8D-219 has two independent rotor systems. The low pressure rotor consists of a single stage fan and a six stage compressor driven by a three stage turbine module. The manufactured Stage 3 and Stage 4 blades were used for these tests. These blades had previously passed the required structural strength, durability (life), and relevant structural qualification tests.

8 Test Location

8.1 Iran Aircraft Industries (IACI) - Company Profile

Iran Aircraft Industries (IACI) was established in 1970 to provide Overhaul and Technical Modifications for aircraft, as well as their associated engines and industrial gas turbines.

Figure 8: Iran Aircraft Industries (IACI) MOA[1] Certificate

[1] Maintenance Organization Approval Certificate

8.2 Test Facility Description

Test-cell is to provide comprehensive performance verification, thrust measurement, and post-validation for turbofan engines. The facility is equipped to handle heavy aero-engines, including the Pratt & Whitney JT8D family (representative of the MD-80 and Boeing 727/737-200 class). This facility integrates advanced Data Acquisition Systems (DAS) capable of real-time performance recording. Core capabilities include:

- Thrust Measurement: Static thrust parameter recording.
- Performance Monitoring: High-resolution logging of Exhaust Gas Temperature (EGT), Fuel Flow, and Compressor/Turbine Speeds (N1/N2).
- Vibration and Structural Analysis: Integrated vibration monitoring and Non-Destructive Testing (NDT) support.

Figure 8-9: Sample of An Engine installed for acceptance test in a test-cell

8.3 Test Set-Up and Special Instructions

8.3.1 Instrumentation and Data Accuracy

Table 8-1- Data accuracy

Parameter	Accuracy	Parameter	Accuracy
TAMB	± 1°F (.555°C)	TT4	± 5°F (2.78°C)
PBAR	± 0.005 psi (0.034 KPA)	PT4	± 0.5 psi (3.4 KPA)
TT2	± 2°F (1.111°C)	PS4	± 0.5 psi (3.4 KPA)
PT2	± 0.01 psi (.1 KPA)	PCP	± 0.5 psi (3.4 KPA)
PS2	± 0.02 psi (.1 KPA)	PB	± 0.5 psi (3.4 KPA)
N1	± 0. 1%	PT6	± 0.5 psi (3.4 KPA)
N2	± 0. 1%	TT7	± 5°F (2.778°C)
TT3	± 3°F (1.665°C)	PT7	± 0.05 psi (0.3KPA)
PT3	± 0.5 psi (3.4 KPA)	Fn	± 46 lb ((20.87 kg)
PS3	± 0.5 psi (3.4 KPA)	Wf	± 0. 1%reading below 4000 pph ± 0.75% reading above 4000 pph

8.3.2 Engine Log Sheet

The following information must be recorded on engine log sheet.

(a) Time of day

(b) Date of test

(c) Weather (during erratic weather conditions record following every half hour).

1 Dry bulb temperature

2 Wet bulb temperature

3 True barometric pressure

(d) Engine serial number

(e) Total time of test run

(f) Grade and specific gravity of fuel

(g) Grade of oil

(h) Reasons for unscheduled shut−downs

Except where noted, following items shall be computed and/or taken and recorded at each power lever position during test.

(a) Time

(b) Observed and corrected N1 and N2 speeds

(c) Observed and corrected thrust (FN)

(d) Observed and corrected fuel flow (WF)

(e) Compressor inlet temperature (TT2)

(f) Compressor inlet pressure (PT2)

(g) Turbine discharge temperature (TT7)

(h)Turbine discharge pressure (PT7). Correct PT7 reading for temperature.

(i) Engine pressure ratio (PT7/PT2)

(j) Fuel inlet pressure

(k) Oil pressure

(l) Breather pressure

(m) Vibration amplitude

(n) Oil consumption

(o) Bleed annulus pressure (PS4)

(p) Inter compressor static pressure (PS3)

(q) Turbine cooling air pressure (PCP)

(f)Vibrations

9 Inspection Requirements

9.1 Pre-Test Inspections

All tests and demonstrations described herein, shall be subject to witnessing by authorized specialists and CVE representatives. At appropriate times prior to testing and during teardown inspections, the engine and its critical components and fourth stage blades on low-pressure turbine in particular, shall be examine to verify their conformity with all contract and specifications requirements under which they were built. After measuring and comparing the dimensions and tolerances of blades and other parts with the drawings, the obtained data should be records for future comparisons.

1- Before start the engine test procedure, do torque inspection check of LPT rotor 4th stage blades (Service Bulletin No., JT8D A6224 revision No.7 date: August 26, 2019)
2- Record LPT blades tip clearances.
3- Record any defect or special conditions of engine or test cell from normal.
4- All necessary inspections of engine, test facility and all procedures necessary to prepare for each test should have been complete before each test.
5- Ensure engine pretest procedures have been completed correctly. If acceptance test is being performed:
    a. All required test tooling is properly installed, secured, or connected to engine.
    b. Inspect engine inlet and exhaust nozzle for foreign objects or damage.
    c. Visually inspect intake and starter gearbox for damage.
    d. Read the engine logbook. Make sure all necessary inspections have been made on work that is complete. Make sure the engine has been released for test.
    e. Make sure the engine lubrication system is correctly serviced.
    f. Make sure the starter is correctly serviced.
    g. Make sure the fuel supply is in correct type, and there is adequate quantity for the test.
    h. Make a check of the basic engine piping
    i. When you operate engines without accessories on the customer pad, make sure the plugs are installed on the gear shafts.
    j. Caution: Examine the engine and test stand thrust and Support Mounts before each engine run to prevent injuries and damage to the engine and test stand.
    k. Make sure the test-cell's thrust and support mounts are serviceable as follows:
        a) Examine the mounts for any crack.
        b) Make sure the mounts are tightly attached.
        c) Make sure the mounts are installed correctly.
    l. Make sure the drain hoses are correctly installed.
    m. Make sure the starter air pipe and leads are correctly installed.
    n. Make sure all electrical connections are tightened correctly.
    o. Make sure vibrations leads, sensors, and instruments are correctly installed.

p. Make sure the throttle operates correctly.
q. Make sure the throttle on the off (zero) position.
r. Make sure the throttle on the idle (38.4 deg.) position.
s. Make sure the shut-off lever on the OFF position.
t. Examine the engine inlet and exhaust for unwanted materials.
u. Make sure the ground area of the test stand is clean.

9.2 During Test inspections

The engine acceptance criteria require that all operating parameters remain within their defined limits throughout testing. A complete list of these limits is provided below. These parameters shall be monitored during the test, and any exceedance of these limits calls for the appropriate troubleshooting procedure to be carried out.

Table 9-1: Engine Operating and Performance Limits (3)

#	Parameter Name	Description	Operating limits	Remark
1	MOP	Main oil pressure	40 – 55 psi (275.8 – 379.2 kpa)	
2	OC	oil consumption	0.10 gallon/hour, 0.4 quarts/hour	
3	MOT	Main oil temperature	275°F (135°C)	329°F (165°C) stabilized idle after snap deceleration from high power running
4	BP	Breather pressure	1.0 psi (6.9 kPa) for O/H engine	1.8 psi (12.4 kPa) for a partially repaired engine
5	EGT	Exhaust gas temperature	Max. T/O 1157°F (625 °C) Nor. T/O 1094°F (590 °C)	

Table 9-2: Maximum Safe Operating Vibration Limits (3)

	Parameter Name	Description	Operating limits
1	VIB 1	Vibration Inlet section (Overall)	Double Amplitude 8.0 mils (0.203 mm)
2	VIB 1	Vibration Inlet section (Low Rotor Frequency)	Double Amplitude 7.0 mils (0.178 mm)
3	VIB 1	Vibration Inlet section (High Rotor frequency)	Double Amplitude 5.0 mils (0.127 mm)
4	VIB 2	Vibration Rear section	Double Amplitude 6.0 mils (0.152 mm)

	Parameter Name	Description	Operating limits
		(Overall)	
5	VIB 2	Vibration Rear section (Low Rotor Frequency)	Double Amplitude 6.0 mils (0.152 mm)
6	VIB 2	Vibration Rear section (High Rotor frequency)	Double Amplitude 4.0 mils (0.102 mm)

Table 9-3: No. 3 Bearing Vibration Rejection Limits (3)

	Parameter Name	Description	Operating limits
1	VIB 1 Or VIB 2	Vibration Inlet Or Rear	Over 2 mils & up to 4 mils Single Amplitude (Up to 5 minutes)
2	VIB 1 Or VIB 2	Vibration Inlet Or Rear	Over 2 mils & up to 4 mils Single Amplitude (Greater than 5 minutes)
3	VIB 1 Or VIB 2	Vibration Inlet Or Rear	Over 4.10 mils & up to 8 mils Single Amplitude (Up to 5 minutes)
4	VIB 1 Or VIB 2	Vibration Inlet & Rear	Above 4.10 mils Single Amplitude (Greater than 5 minutes)
5	VIB No. 1 & No. 2 VIB 1 Or VIB 2	Vibration Inlet & Rear	Above 8 mils Single Amplitude (Any Time)

Table 9-4- Acceptance Vibration Limits Amplitude (3)

	Parameter Name	Description	Operating limits	Remark
1	VIB 1	Vibration Inlet section (Overall)	Double Amplitude 3.0 mils (0.076 mm)	
2	VIB 2	Vibration Rear section (Overall)	Double Amplitude 3.0 mils (0.076 mm)	

Table 9-5- JT8D-219 Power Setting (3)

Power Setting	Corr. Thrust LBS (Kg)	Corr. N1 Speed RPM	Corr. N2 Speed RPM	Corr. TSFC LBS/HR/LB (Kg/HR/Kg)	Max. Allowable Observed N1 Speed RPM (%)	Max. Allowable Observed N2 Speed RPM (%)
Max. Take-Off	21700 (9841)	7860	11760	-	8350(101.6)	12550(102.5)
Normal Take-Off	21000 (9524)	7670	11680	0.515	8120(98.8)	12350(100.9)
Max. Climb	18900 (8573)	7240	11440	0.489	-	-
Max. Cruise	16400 (7439)	6890	11170	0.481	-	-

Table 9-6- Engine Fuel and Oil Leakage Limit (3)

Location	Fluid	Allowable leakage	Remark
Gearbox starter drive overboard drain	Oil	10 CC/HR	

Location	Fluid	Allowable leakage	Remark
Gearbox Hyd. pump drive overboard drain	Oil	10 CC/HR	
No. 1, 2, 3 bearings fluid seal drain	Oil	0.5 CC/Minute (10 Drops/Minute) for each drain	
No. 4 bearing Air check valve drain	Oil	Per SB:5989 Oil leakage at idle power is normal Post SB:5989 No Oil leakage allowable	
Fuel pump drive overboard drain	Oil	10 CC/HR	
Fuel pump drain	Fuel	60 CC/HR with engine running and/or shut down	
Fuel Control drain	Fuel	Noun	
P & D valve	Fuel	Noun	
Exhaust Case No. 6 Bearing sump	Oil	Oil wetness not resulting in oil puddling within 20 minutes after engine shutdown	
Combustion chamber Drain and/ or Wet stained or streaked 1th stage turbine vanes/blades	Oil	* **	See Notes

*** NOTE:** Oil wetting on the first stage turbine vanes/blades sometimes can be described as an irregular radial pattern and streaks outward. Such indications if seen, can incorrectly be reported as cracks.

**** NOTE:** Oil leakage from the Combustion Chamber Drain and/or Wet 1st Stage Turbine Vanes/Blades is not permitted if the engine is Post SB A6196 – Improved No.5 Bearing Oil Return and Compartment Sealing.

The acceptance thrust graph shown in Figure 9-1. The performance parameters are corrected for environmental conditions, and the effects of the test cell, instruments, and the rest of the engine are constant during the tests. Therefore, systematic errors and environmental factors are eliminated, and by comparing the results of the engine test with the MAPNA newly manufactured blades to the results of the engine test with OEM parts, one can be confident about the effect of the new blades on engine performance across all engine operating rates.

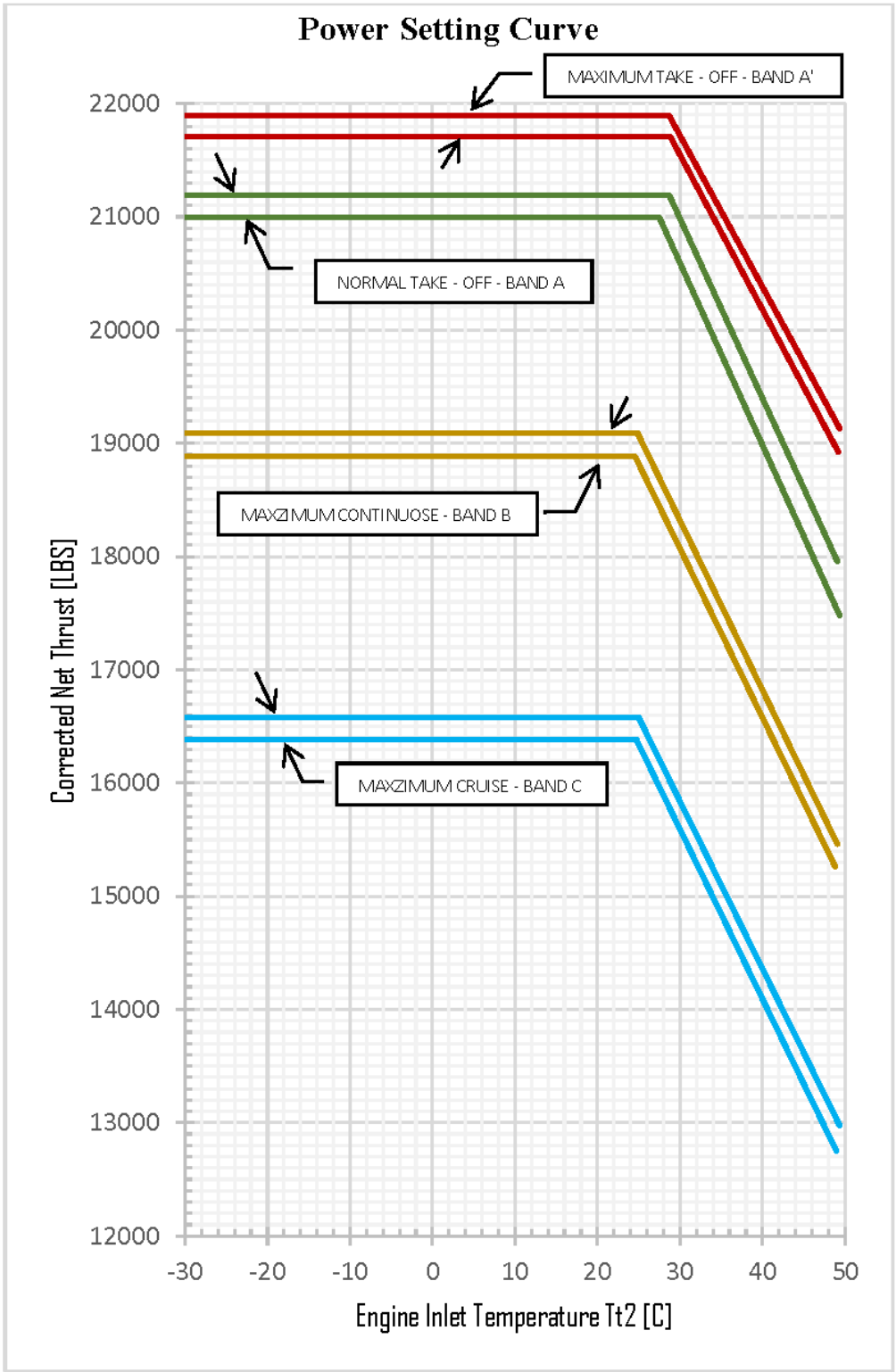

Figure 9-1: Power Setting Curve (4)

9.2.1 Post-Test Inspections

Upon completion of the total tests run, the engine shall be disassembled sufficiently to allow a detailed inspection of all vital working parts specifically low-pressure turbine third or fourth stage blades.

When the engine completing the whole tests, LPT blades shall disassemble for examination. Prior to cleaning, the engine parts shall be given a "Dirty Inspection" to inspect for evidence of leakage, oil coking, unusual heat patterns, excessive corrosion and other abnormal conditions. Any evidence should be record with high resolution picture. Following the "Dirty Inspection", the engine parts shall be cleaned and measurements shall be taken as necessary to inspect for excessive wear or distortion. Then all parts shall be subjected to a "Clean Inspection" consisting of a visual examination, NDT and condition assessment.

1- Upon completion of the engine test run procedure and before engine disassemble, do the torque inspection check of LPT rotor 4th stage blades (AD: 2021-22-19 and Service Bulletin No., JT8D A6224 revision No.7 date: August 26, 2019)
2- Also, do inspection check of LPT 4th stage disk and blades as per JT8D EM (P/N: 822504) section 72-53-04, (Turbine disk and blades assembly, stage 4- inspection-01)
3- Perform the Fluorescent Penetrant Inspection (FPI) according to 72-53-00-23-000.

9.3 B-2 Test: 3rd and 4th stage

Upon successful validation of the LPT blade fourth stage, the manufactured LPT blade third stage will replace the OEM Stage 3rd parts. The test will then proceed with both Stage 3rd and Stage 4th manufactured blades installed, maintaining the same quality standards and replicating the additional inspection steps previously performed during the Stage 4 test campaign for the Low Pressure Turbine (LPT 3&4) stages.

After achieving the goals in section 9-1 and pass the tests of LPT blades fourth stage, time to do the same process in test procedures and post-test for stage 3rd and 4th together. This part is the same as the section 9-2 (4th stage) and the pre-inspections, test procedures and post –inspections would be done the same, also do inspection check of LPT 3rd stage disk and blades as per JT8D EM (P/N: 773128) section 72-53-03, (Turbine disk and blades assembly stage 3- inspection-01)

10 References

1. **Title 14 - Aeronautics and Space Chapter I - Department of Transportation Subchapter C - Aircraft.** ***s.l. : Federal Aviation Administration, 2025. Part 33 Airworthiness Standards: Aircraft Engines.***

2.* JT8D ENGINE MANUAL (PN 773128) ENGINE TESTING 01 to 09.** ***s.l. : Pratt & Whitney.

3.* Table 1302 of JT8D ESM.** ***2015. 72.00.00.

4.* JT8D ENGINE MANUAL (PN 773128) 72−00−00 TESTING−06 Figure 1304.** ***s.l. : Pratt & Whitney, 2005.

Table 10-1- Approval of Test Plan

Roles	Signature/Date
Prepared by: DE [Vahid Aramian] [Iman Laraibi] [Aghil FazelNiari]	
Provided Proof of Compliance by: DOM [Golsa Ghanati]	
Checked & Approved by: CVE [Golsa Ghanati] [Alireza Navaei] [Ali Farokhipour]	
Checked & Released by: HOA [Alireza Taghipour]	